Galaba Vamsi

Machine Learning Engineer, Intern / New Grad (graduating May 2027)
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SUMMARY

Mechatronics at IIT Bhilai. Applied ML that I test on the real thing: a voice-clone detector checked on live phone calls (top 10 of 4,500+ teams at RAKSHAM) and 5th of 232 teams in Kaggle's WEAR challenge. Co-author of an embodied-AI evaluation paper (CoRL 2026 workshop).

EDUCATION

B.Tech. in Mechatronics, Indian Institute of Technology (IIT) Bhilai Aug 2023 - May 2027
Coursework: AI and ML, Deep Learning for Computer Vision, Digital Image Processing, Optimal Control, DSA, OOP.

PROJECTS

Pehchaan (team of 2; Top 10, RAKSHAM (IIT Delhi, I4C and Amazon), 2026) Voice-clone payment-fraud detection for UPI; I led ML and backend. github.com/Galabavamsi/pehchaan

- Fine-tuned XLS-R-300M + AASIST on consented Telugu, Hindi, and English speech plus clones from 13 TTS and voice-conversion generators: 4.8% equal error rate (EER) vs 34.3% off the shelf; 6.0% on 6 unseen generators (pre-registered splits, 19k test clips).
- Found a sim-to-real distribution shift: 24.6% EER on 48 live carrier calls. Modelled the speakerphone channel and retrained: 6.8% EER, clones caught 48% to 75%, 0% of genuine callers flagged. The detector stays a non-deciding signal until a 50-call, 5%-false-flag gate passes.

WEAR Dataset Challenge @HASCA 2026 (Kaggle, 2026) Exercise recognition from wearable IMU and video tiles of four unseen subjects: 5th of 232 teams (macro-F1 0.93137, private leaderboard). Built on a public notebook; per-subject whitening, ZCA video input, and a multi-run ensemble lifted my reproduction from 0.92497 to 0.93825 public. Rejected a global whitening that raised cross-validation but lowered the leaderboard (a cross-subject leak).

World Model on a Laptop (Push-T, 2026) Reproduced the LeWorldModel (JEPA) recipe at 1.9M parameters on an 8 GB GPU; rollouts fell behind a frozen-latent baseline after about 2 steps and the planner scored 0, all reported in a public write-up.

Amazon ML Challenge (team, 2025 and 2026) Product price prediction with a Qwen2-VL 7B vision-language model and a ViT/RoBERTa two-tower model (2025, All India Rank 700); All India Rank 200 in 2026.

Arista Wi-Fi RRM (team, Inter IIT Tech Meet 14.0, 5th place, Dec 2025) I/Q interference classification (STFT, DBSCAN, Dual-CNN; 16 classes) for client-aware radio resource management in NS-3.

EXPERIENCE

Co-founder, Robotropic Physical-AI startup, with Varun Rao | robotropic.com Oct 2026 - Present

- Building Harness, a workspace where a factory's team and AI agents change robot skills safely: each change is checked against safety limits, rolled out cell by cell, and recorded. I lead the simulated factory floor and the change gateway.

Research Intern, IIT Bhilai Advisor: Dr. Arzad Alam Kherani | COMET Foundation project May 2025 - Jul 2025

- Built RIS-SIM, an open-source Python emulator for reconfigurable intelligent surface (RIS) radio links with USRP-style SDR interfaces; designed RIS unit cells in CST Studio Suite (IEEE INDICON 2025, co-author). Rebuilt it in 2026 as RIS-SIM v2 (103 tests), then as sremu.

Co-founder, Human Slop Anti-AI social app, with Varun Rao | humanslop.in Mar 2026 - Present

- Expo (React Native) app for web and Android, Fastify API, PostgreSQL on Supabase; live on the web and Google Play (50+ downloads).

OPEN-SOURCE CONTRIBUTIONS

Inspect Robots, Robocurve (YC S26) github.com/robocurve/inspect-robots Sep 2026 - Oct 2026

- Evaluation framework for embodied-AI policies. 2 merged PRs: a configurable LLM-API retry policy across 5 wire protocols (PR #476) and unknown git state no longer logged as clean (PR #475); 3 open, including checkpoint/resume for long eval sets (PR #514).
- Wrote and maintain inspect-robots-wandb, a Weights & Biases plugin (pypi.org/project/inspect-robots-wandb): 100% coverage gate, CI on Python 3.10-3.13, trusted publishing.

HFlow, Hebbian Robotics (YC S26) github.com/Hebbian-Robotics/hflow Sep 2026 - Oct 2026

- 8 merged PRs (Python, Apache Arrow) to an SDK that verifies robot training data: LeRobot and robot-log conversion now keeps data whole or refuses it with a reason. Blocked silent 12- to 10-bit depth loss (PR #400), typed Arrow columns from every message (PR #657), fixed video-cache collisions (PR #353).

PUBLICATIONS AND PREPRINTS

- **Co-author**, *Inspect Robots: Evaluating the Capabilities and Safety of Embodied AI*, arXiv:2610.06306; accepted to the SPAIS Workshop at CoRL 2026 (poster).
- **Co-author**, 2 RIS papers: IEEE INDICON 2025 (RF energy-harvesting RIS) and *Ad Hoc Networks* (Elsevier), 2026 (systematic review of RIS simulation models).
- **Sole author**, *Timed RIS Control in the Loop*, preprint draft, arXiv submission pending (github.com/Galabavamsi/sremu-paper).

SKILLS

ML: PyTorch, fine-tuning (speech, vision-language), evaluation design (EER, held-out generators, cross-validation, leakage checks), ensembling, int8 inference, Weights & Biases, LightGBM
Languages and serving: Python, SQL, TypeScript, FastAPI, Docker, Git, Linux, GitHub Actions, pytest
AI-assisted development: Claude Code; I set the design and verify results with tests and independent references.

ACHIEVEMENTS

- **Top 10 (Grand Jury round) of 4,500+ teams nationwide**, RAKSHAM AI Cybersecurity Hackathon (IIT Delhi, I4C and Amazon), Oct 2026; Pehchaan, team of 2.
- **5th of 232 teams**, WEAR Dataset Challenge @HASCA 2026 (Kaggle); **All India Rank 200** (2026) and **700** (2025), Amazon ML Challenge (team); **5th place**, Arista Networks Wi-Fi challenge, Inter IIT Tech Meet 14.0 (team).
- Prime Minister's Scholarship Scheme (PMSS) scholar, 2024; Subroto Mukherjee Scholarship (IAFBA), 2023; Top 1% in JEE Advanced; attended YC Startup School, Bengaluru.